Lecture 1: Binomial Asset Pricing Model

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1 Introduction

In finance, **long** means buying a security such as a stock, commodity or currency, with the expectation that the asset will rise in value. In mathematics, we use a positive number to represent a long position. For example, $+1$ share means long one share.

Short (selling) means borrowing a security from others and selling it[1], with the expectation that the asset will drop in value, so that the borrower can purchase it back with a lower price. In mathematics, we use a negative number to represent a short position. For example, -1 share means short one share.

A **derivative security** is a financial instrument whose price is dependent on one or a number of underlying financial assets (e.g. stocks). In itself, the derivative security is no more than an agreement between two contracted parties to buy or sell an asset at a fixed price on or before a date of expiration. We call the above-mentioned price and the date of expiration strike price and maturity respectively. Examples of derivative securities: forward contract, European call/put option.

A **forward contract** is a non-standardized contract between two parties to buy or sell an asset at a specified future time (maturity T) at a price (strike K) agreed today (time 0). The value of the contract at time T is $S_T - K$, where S_T is the asset price at time T.

An **option** is similar as a forward, but an option contract just offers the buyer of the option the right, but not the obligation to buy or sell the underlying asset at a pre-specified strike price K on or before maturity T.

An **European call option** is one kind of option that gives the buyer of the option the right, but not the obligation to buy the underlying assets at a strike price K on maturity T. The value of the contract at T is $(S_T - K)^+ = \max(S_T - K, 0)$, where S_T is the asset price at time T.

An **European put option** is one kind of option that gives the buyer of the option the right, but not the obligation to sell the underlying assets at a strike price K on maturity T. The value of the contract at T is $(K - S_T)^+ = \max(K - S_T, 0)$, where S_T is the asset price at time T.

Note that, we have a **put-call parity**:

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K. \tag{1}$$

In words, the time-T value of a call less the time-T value of a put (with same strike/maturity) is equal to the time-T value of a forward contract (with same strike/maturity).

How to give a fair price of these securities before maturity? In what sense is it fair?

To analysis it, one should first rule out arbitrage. An **arbitrage** is a trading strategy which begins with no money, has a zero probability of losing money, and has positive probability of making money (most unfair thing, uhh?). An arbitrage may exist in the real market, but it is necessarily fleeting, as smart traders discover it, they quickly take advantage of it and remove it.

We will assume no arbitrage henceforth. And the pricing methodology under this assumption is called no-arbitrage pricing. The first example to illustrate this methodology, is a binomial model.

2 One-Period binomial model

Without loss of generality, we may assume that $0 < d < u$ and $0 < p < 1$. The simplest one period binomial financial market consists of a stock S evolving according to Figure 1, and a money market with risk-free rate

[1] In practice, short selling requires the borrower to have certain balance in his account initially, and requires him to pay interest rate to the lender, but we do not need to worry about the details here.

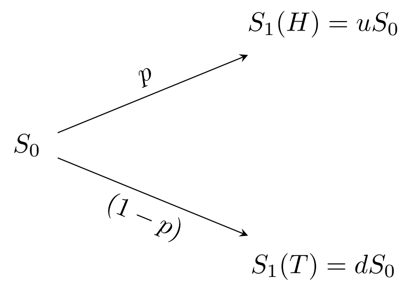

Figure 1: Coin toss and binomial model.

r. To avoid arbitrage, one should have

$$0 < d < 1 + r < u. \tag{2}$$

For otherwise, say, $0 < d < u < 1 + r$, then an investor with no money can short 1 share of the stock, and deposit the proceeds (S_0) into the money market. At time 1, with 100% probability, the investor is able to buy to close the short position and still has a profit. This is an arbitrage.

Under no arbitrage, let us derive a fair value of a call option with strike K and maturity at time 1. The idea is replication: if we are able to replicate the payoff of the call option along any possible paths (any coin tosses) with a portfolio of stock and cash, then the initial cost of setting up this portfolio must be the same as the initial value of this call option.

Example 1. Assuming that $S_0 = \$4, u = 2$, and $d = \frac{1}{2}$ in Figure 1, and interest rate $r = \frac{1}{4}$. Consider a call option with strike at 5 and maturity at time 1. The payoff of this option is either 3 (if a head shows up) or 0 (if a tail shows up). On the other hand, consider a portfolio consists of borrowing 0.8 from money market and buying 0.5 shares of stock. At time 1, we have to pay back $\$0.8 \times (1 + \frac{1}{4}) = \$1$ to the money market. If a head shows up at time 1, then the total value of the portfolio at time 1 is $0.5 \times \$8 - \$1(\text{debt}) = \$3$; if a tail shows up, the total value of the portfolio at time 1 is $0.5 \times \$2 - \$1(\text{debt}) = \$0$. In all cases, the value of the portfolio matches the payoff of the call option at time 1.

The arbitrage-free price of the call option should be equal to the initial cost of the portfolio, \$1.20. For otherwise, say, the option is sold at \$1.21, then we could short one option, and with the proceeds (\$1.21) to setup the above portfolio and deposit the extra \$0.01 to the money market. At time 1, the above portfolio can be used to close the short position of the call option, and the 1 cent deposit will give us \$0.0125. This is an arbitrage.

From this example we know that, to price a derivative security, it is sufficient to setup a portfolio of stocks and cash to replicate the payoff of this derivative when it matures. In an one-period binomial market, we can denote by $V_1(H)$ and $V_1(T)$ the payoff[2] of a derivative security at time 1. We seek a portfolio with Δ_0 shares of stock and total initial value X_0, to replicate the payoff of the derivative security along all possible paths (all possible coin tosses). Mathematically, solve for Δ_0 and X_0 in the following system:

$$\begin{aligned} \Delta_0 S_1(H) + (1+r)(X_0 - \Delta_0 S_0) &= V_1(H), & (3)\\ \Delta_0 S_1(T) + (1+r)(X_0 - \Delta_0 S_0) &= V_1(T). & (4) \end{aligned}$$

First, it is straightforward to obtain

$$\Delta_0 = \frac{V_1(H) - V_1(T)}{S_1(H) - S_1(T)}. \tag{5}$$

Second, multiply (3) by a constant $\tilde{p}$, and (4) by a constant $\tilde{q} = 1 - \tilde{p}$, then add them. After some easy algebra, we get

$$X_0 + \Delta_0 \left(\frac{1}{1+r} [\tilde{p} S_1(H) + \tilde{q} S_1(T)] - S_0 \right) = \frac{1}{1+r} [\tilde{p} V_1(H) + \tilde{q} V_1(T)]. \tag{6}$$

[2] $V_1 = (S_1 - K)^+$ if it is an European call, $V_1 = (K - S_T)^+$ if it is an European put, etc.

If we choose $\tilde{p}$ so that[3]

$$S_0 = \frac{1}{1+r}[\tilde{p}S_1(H) + \tilde{q}S_1(T)], \tag{7}$$

then we obtain the initial cost of the portfolio

$$X_0 = \frac{1}{1+r}[\tilde{p}V_1(H) + \tilde{q}V_1(T)]. \tag{8}$$

Remark 1 *In an arbitrage-free one-period binomial market, we have arguments showing that pricing is equivalent to replicating. In solving the replicating problem, we obtain Δ_0 - the delta hedge coefficient, and the arbitrage-free price of the derivative, X_0. The delta hedge coefficient has a nice formula,*

$$\Delta_0 = \frac{\Delta V_1}{\Delta S_1}. \tag{9}$$

That is, the rate of change of the derivative price with respect to the price change of the underlying. Formula (8) for X_0 has an intuitive interpretation. Note that $\tilde{p} \in (0,1)$[4], and it can be perceived as a probability of head. This probability is not the actual (physical) probability[5], but rather the **risk-neutral** *probability. Under the risk-neutral probability, the expected growth of the stock price is equal to the risk-free rate. And under the risk-neutral probability, the arbitrage-free price of the derivative, is equal to the discounted expectation of the payoff of the derivative. This result highly simplifies pricing.*

Remark 2 *There are some practical issues of the hedging/replication:*

- *shares of stock can be subdivided for sale or purchase - not a big deal if the volume if large,*
- *the interest rate for investing is the same as the interest rate for borrowing - close to true for large institutions,*
- *zero bid-ask spread - maybe true in practice if the stock is highly liquid,*
- *at any time, the stock price can only take two possible values in the next period - not true in practice, but in some situations, it can work remarkably well with improvements (multi-period, trinomial tree, etc)*

[3] In fact, $\tilde{p} = \frac{1+r-d}{u-d}$ will do the job.

[4] Try to figure out why it is so.

[5] For the actual probability p, it is usually the case that $S_0 < \frac{1}{1+r}[pV_1(H) + qV_1(T)]$. Otherwise, we'd better invest in the money market to earn risk-free rate.